DA3452 Speech and Language Technology

Module 4 Test

Transformers

Duration 60 minutes Maximum marks 60 Questions 31

Name:	Roll No.:	Date:

Instructions

- Section A: 15 MCQs, 27 marks. Section B: 7 MSQs, 14 marks. Section C: 9 fill-in-the-blank questions, 19 marks. All questions are compulsory; marks are shown against each question.
- MCQ: Select one option.
- MSQ: Select all correct options. An MSQ may have one or more correct options.
- MSQ marking: Selecting any incorrect option gives 0 for that question. Otherwise, marks are awarded in proportion to the correct options selected (rounded to the nearest 0.5). There is no negative marking.
- Token positions are numbered from 1 unless a question states otherwise.
- **Malpractice can result in disciplinary actions as per the Senate mandate.**

SECTION A MCQ

Questions 1-15 | 27 marks

1. [1 mark] A decoder-only Transformer is trained to generate navigation instructions. One training example uses the input sequence [<START>, turn, left] and the shifted target sequence [turn, left, <EOS>].

At the input position containing turn, which input tokens can be attended to under ordinary causal self-attention?

A. <START>
B. <START>, turn, left
C. <START>, turn
D. turn, left

2. [1 mark] A Transformer processes 20 tokens from a support ticket. Its input matrix X has shape 20×24. It has three attention heads, each with $d_k = d_v = 8$. For one head, $Q = XW^Q$.

What are the shapes of W^Q and QK^T, respectively?

A. 24×8 and 20×20
B. 20×8 and 20×20
C. 24×8 and 20×8
D. 24×8 and 8×8

3. [1 mark] Consider $x = (1, 3)$ and $y = (2, 6)$. Apply LayerNorm and RMSNorm separately to each vector, using unit gains and zero offsets. Ignore ε. Compare the normalized outputs before any residual addition.

How do the outputs for x and y compare under the two normalization methods?

A. They are identical under both LayerNorm and RMSNorm.
B. They differ under LayerNorm but are identical under RMSNorm.
C. They differ under both LayerNorm and RMSNorm.
D. They are identical under LayerNorm but differ under RMSNorm.

4. [**2 marks**] A two-block decoder-only Transformer is trained for email completion using the complete gold input sequence and shifted next-token targets. The first block uses unmasked self-attention; the second block uses causal self-attention.

An engineer claims that the causal mask in the final block is sufficient to prevent future-token leakage. Which statement is correct?

A. The final mask is sufficient because states at future positions are excluded from every final-block query.
B. Leakage can remain because permitted first-block states may already contain future-token information.
C. The final mask is sufficient because normalization in each block removes any future-token information.
D. Leakage can remain because the shared feed-forward network transfers information across positions.

5. [**2 marks**] A causal language model is trained to complete equipment records. The loss is computed only on completion targets, and predictions in the completion can attend to the prefix. The token pump appears in the prefix but not in the completion. The input embeddings and output projection are untied, and there is no regularization.

Which statement is correct about the gradient of the pump input embedding?

A. It must be zero because the prefix positions are excluded from the loss and receive no gradient.
B. It can be nonzero only if pump also occurs among the completion targets.
C. It can be nonzero because completion predictions can depend on the prefix through attention.
D. It must be nonzero because every completion prediction depends on every prefix token.

6. [**2 marks**] A decoder-only Transformer classifies sentiment using a head applied to the final input token's final-layer representation. Consider these two token sequences, with each displayed item treated as one token:

[this, film, is, good]
[this, film, isn't, good]

A bug replaces the attention output with the zero vector in every block. Token and position embeddings, feed-forward networks, normalization and residual connections still operate. Parameters are fixed and dropout is disabled.

Which statement must hold for the predicted class probability distributions?

A. They can differ because the position embeddings let the final position reconstruct the preceding tokens.
B. They must match because the final-position inputs are identical and every remaining operation is position-wise.
C. They can differ because the shared feed-forward network combines information carried from preceding tokens.
D. They must match because LayerNorm maps every position's representation to one identical normalized vector.

7. [**2 marks**] A pre-normalized feed-forward residual branch, without biases, is defined by $a = \mathrm{ReLU}(\mathrm{LN}(x)W_1)$ and $h = x + aW_2$.

For a single-token loss, both a and the incoming gradient $g = \partial L/\partial h$ are nonzero, but $W_2 = 0$.

What are the gradients with respect to x, W_1 and W_2, respectively?

A. g; zero; zero
B. zero; zero; nonzero
C. g; nonzero; nonzero
D. g; zero; nonzero

8. [2 marks] A decoder uses learned absolute position embeddings. The sequence [open, the, door] is processed alone with position IDs 0, 1, 2. It is also processed as [PAD, PAD, open, the, door]. Parameters are fixed and dropout is disabled.

A padding mask, when enabled, excludes PAD keys in every block. PAD outputs are ignored.

Which setup guarantees the same next-token probability distribution after door for arbitrary model parameters?

A. Use position IDs 2, 3, 4 for the three real tokens and enable the padding mask.
B. Use position IDs 0, 1, 2 for the three real tokens and enable the padding mask.
C. Use position IDs 0, 0, 0 for the three real tokens and enable the padding mask.
D. Use position IDs 0, 1, 2 for the three real tokens and disable the padding mask.

9. [2 marks] Two equal-length histories differ in their earlier tokens but have identical final four tokens and identical position IDs. The same two-block causal decoder originally processed each whole history. Parameters are fixed, dropout is disabled, and there is no external memory.

Before processing an identical new token, compare two trimming methods:

R: Clear all caches and recompute the final four tokens from scratch.
C: Delete the earlier cache entries, but retain the originally computed keys and values for the final four tokens in every block.

Position IDs are unchanged in both methods. How do the next-token probability distributions compare across the two histories under R and C, respectively?

A. They must be identical under R; they can differ under C.
B. They must be identical under both R and C.
C. They can differ under R; they must be identical under C.
D. They can differ under both R and C.

10. [2 marks] Two attention heads produce row vectors h_1 and h_2, each of shape 1×2. Their combined output is $y = [h_1\ h_2]W^O$, where W^O has shape 4×4.

The concatenation order is changed to $[h_2\ h_1]$. Which change to W^O guarantees the original output y for arbitrary head outputs and arbitrary original W^O?

A. Swap the first two columns with the last two columns.
B. Swap the first two rows with the last two rows.
C. Transpose W^O.
D. Leave W^O unchanged.

11. [2 marks] A subject-line generator has three Transformer blocks. Positions within a block can be processed in parallel, but the blocks must be applied successively.

Compare one teacher-forced training forward pass, with all input tokens available, to autoregressive generation of four new tokens. In the generation case, the prompt has not yet been processed, and the fourth new token is not fed back into the model. Ignore embeddings, the language-model head and token selection.

How many sequential Transformer-block stages are required in the training and generation cases, respectively?

A. 3 and 3
B. 12 and 3
C. 3 and 12
D. 12 and 12

12. **[2 marks]** A robot generates exactly two command tokens, followed by <EOS> with probability 1. The token probabilities are given below.

Step / prefix	Token probabilities
First token	move: 0.60; turn: 0.40
After move	left: 0.60; right: 0.40
After turn	left: 0.95; right: 0.05

Which pair gives the sequence selected by greedy decoding and the highest-probability complete sequence, respectively?

A. move left; move left
B. turn left; turn left
C. turn left; move left
D. move left; turn left

13. **[2 marks]** A newspaper headline generator applies temperature scaling, retains the top three tokens, renormalizes and samples. At a particular step, the article, partial headline and logits are fixed. The vocabulary has more than three tokens, and all logits are distinct.

The temperature is increased from 0.5 to 1.0. What happens to the retained token identities and the renormalized probability of the highest-logit token?

A. The retained identities stay the same; the probability decreases.
B. The retained identities stay the same; the probability increases.
C. The retained identities change; the probability decreases.
D. The retained identities change; the probability increases.

14. **[2 marks]** A generator describes a ceramic mug. After the prefix "This mug has a", its simplified next-token distribution at temperature $\tau = 1$ is glossy: 0.50, matte: 0.30, rough: 0.20. Both methods below use temperature $\tau = 0.5$ and a top-p threshold of 0.60.

Method 1 applies temperature scaling and then top-p filtering. Method 2 applies top-p filtering first and then temperature scaling within the retained set.

Which tokens remain available for sampling under Method 1 and Method 2, respectively?

A. glossy; glossy
B. glossy and matte; glossy
C. glossy; glossy and matte
D. glossy and matte; glossy and matte

15. **[2 marks]** A repair assistant generates exactly two tokens, followed by <EOS> with probability 1. First-token probabilities are inspect: 0.40, replace: 0.35, reset: 0.25. Second-token probabilities are given below. Beam search uses width 2 and cumulative model probability, with no length normalization.

Prefix	now	later	again
After inspect	0.40	0.30	0.30
After replace	0.40	0.35	0.25
After reset	0.90	0.05	0.05

Which pair gives the beam-search output and the highest-probability complete sequence, respectively?

A. inspect now; inspect now
B. reset now; reset now
C. reset now; inspect now
D. inspect now; reset now

SECTION B MSQ

Questions 16-22 | 14 marks

16. [2 marks] A Transformer feed-forward branch, without biases, is $F(x) = \mathrm{ReLU}(xW_1)W_2$, where W_1 has shape $d \times 4d$ and W_2 has shape $4d \times d$.

An engineer removes ReLU from every feed-forward branch but continues to train both matrices. Attention, normalization and residual connections are unchanged. Which statements are correct?

A. Each branch's trainable parameter count falls from $8d^2$ to d^2.
B. For fixed weights, each branch is equivalent to a single $d \times d$ linear map.
C. The Transformer as a whole can still be nonlinear in its inputs.
D. The feed-forward branch now mixes information across different token positions in the sequence.

17. [2 marks] An attention sublayer receives X of shape $N \times 12$. It has three heads, each with $d_k = 4$ and $d_v = 6$. Head outputs are concatenated and multiplied by an output matrix W^O before residual addition and a feed-forward sublayer. The residual-stream width is 12.

Which statements are correct?

A. Each head output has shape $N \times 6$.
B. W^O can have shape 18×12.
C. Multiplying attention weights by V requires $d_v = d_k$.
D. The concatenated head output has shape $N \times 18$.

18. [2 marks] An attention head has values $v_1 = (1, 0)$, $v_2 = (0, 1)$ and $v_3 = (0.5, 0.5)$. Compare attention-weight vectors $\alpha = (0.2, 0.2, 0.6)$ and $\beta = (0.45, 0.45, 0.1)$. Contexts are measured before the output projection. When values are changed, queries and keys stay fixed.

Which statements are correct?

A. The initial contexts for α and β are identical.
B. For these three values, the context uniquely determines the attention weights that produced it.
C. Adding the same nonzero δ only to v_3 changes the α context six times as much as the β context.
D. Applying the same output projection to the two initial contexts must produce identical outputs.

19. [2 marks] A faulty causal-attention implementation computes softmax over all positions, including future positions. It then sets future-position weights to zero after softmax, does not renormalize, and forms the weighted sum of values.

There is at least one allowed and one future position, and all scores are finite. The query is fixed. Projected keys and values can be changed independently, and the context is measured before the output projection.

Which statements are correct?

A. The remaining attention weights sum to 1.
B. The context can depend on future keys despite their weights being set to zero.
C. The context can depend on future values despite their weights being set to zero.
D. Renormalizing the remaining weights recovers the correct masked-attention weights.

20. [**2 marks**] The same Transformer processes either 64 or 128 unpadded tokens. Model width, number of heads, number of blocks and batch size are fixed. Its position-embedding table was allocated for 1,024 positions. Each head uses dense attention and stores a full $N \times N$ score matrix.

When the sequence length increases from 64 to 128, which statements are correct?

A. The total number of trainable parameters doubles.
B. The number of stored attention-score entries quadruples.
C. The number of scalar multiplications in feed-forward projections quadruples.
D. The number of scalar multiplications in the entire attention computation, including Q, K, V and output projections, quadruples.

21. [**2 marks**] In a pre-LayerNorm decoder, each attention and feed-forward sublayer receives the LayerNorm of its residual-stream input and adds its output back to that stream. A final LayerNorm is applied before the language-model head.

Let x_i be the initial residual-stream vector after token and position embeddings. Compare the original inputs with $x'_i = x_i + c_i\mathbf{1}$, where c_i is a scalar that can vary by position, $\mathbf{1}$ is the all-ones vector, and at least one c_i is nonzero. Parameters and masks are fixed, and dropout is disabled.

Which statements are guaranteed?

A. Corresponding heads have identical attention weights in the two runs.
B. Residual-stream vectors are identical in the two runs after the first block.
C. Final vocabulary probability distributions are identical at each position.
D. The same guarantees hold if every LayerNorm is replaced by RMSNorm.

22. [**2 marks**] A scalar attention head has $d_k = d_v = 1$, query $q = 1$, keys $k_1 = k_2 = 0$, and values $v_1 = 0$ and $v_2 = 2$. It uses ordinary softmax attention. Let c be its context and $L = \frac{1}{2}c^2$. Treat q, k_1, k_2, v_1 and v_2 as independent variables.

Which statements hold at these values?

A. $\partial L/\partial q = 0$.
B. Both key gradients are zero.
C. Both value gradients equal 1.
D. Increasing k_1 while holding all other variables fixed increases the loss.

SECTION C Fill in the blanks

Questions 23-31 | 19 marks

23. **[1 mark]** In scaled dot-product attention, $d_k = 16$ and the query-key dot product is 12.
The scaled attention score is _____.

24. **[1 mark]** A Transformer processes a 12-token medical report. Within one Transformer block, the position-wise feed-forward network uses matrices W_1 and W_2 that are shared across positions.
Within that block, the number of distinct matrix pairs (W_1, W_2) used across these 12 positions is _____.

25. **[2 marks]** An attention head uses softmax attention over two retrieved records, A and B, with weights 0.4 and 0.6. Their values are A: (1, 0) and B: (0, 1). An exact duplicate of A is added, with the same projected key and value as A.
The query and all existing projected vectors stay unchanged. All three records are accessible, and there are no position-dependent score terms.
The new context vector, before the output projection, is _____.

26. **[2 marks]** A decoder processes 12 tokens through three blocks. In each block, position j may attend only to positions $j - 2$, $j - 1$ and j, when those positions exist. Feed-forward networks, normalization and residual connections act separately at each position. There is no other mixing across positions.
The earliest original input position whose information can reach the final representation at position 10 is position _____. Use position numbers 1 through 12.

27. **[2 marks]** A decoder has two blocks and two heads per block, with $d_k = d_v = 4$ in each head. Its key/value cache stores all processed tokens in every head of every block.
A six-token prompt has already been processed and cached. The decoder generates three new tokens by ordinary autoregressive decoding. Immediately after the third new token is selected, before it is fed into the model, the cache contains _____ scalar entries for this one sequence. Count keys and values only.

28. **[2 marks]** A decoder has six blocks, model width 8, one attention head with $d_k = d_v = 8$, and a conventional feed-forward hidden width of 32. Initially, each block has independent attention and feed-forward matrices.
All six blocks are changed to share one set of attention and feed-forward matrices, while still performing all six successive block applications. Count only the Q, K, V and output projection matrices and the two feed-forward matrices; ignore biases.
This change saves _____ trainable weights, and the model still performs _____ successive block applications.

29. **[3 marks]** A three-block decoder processes [beep, beep, beep, beep]. It has no positional embeddings or position-dependent score terms; the stated causal mask remains. Ordinary causal attention includes the current position. Feed-forward networks, normalization and residual connections act separately at each position; their parameters are shared across positions. Dropout is disabled, and the same language-model head is applied at every position.
The number of distinct next-token probability distributions produced across the four positions is _____.

30. **[3 marks]** A thermostat language model has exactly three vocabulary tokens. Their embedding rows are heat: (2, 0), cool: (0, 2) and hold: (1, 1). The output projection is tied to the embedding matrix: $u = hE^T$. There is no output bias, and full-vocabulary softmax is used at temperature 1.

At a particular step, P(heat) = 4 P(cool). The value of P(hold) is _____.

31. **[3 marks]** A language model ties its input embedding matrix E to its output projection, so that logits are hE^T. It uses full-vocabulary softmax at temperature 1 and a single cross-entropy loss. There is no auxiliary loss or regularization.

The token valve appears twice in the prefix but is not the target. Gradients arriving at the two embedding-lookup outputs are $g_1 = (-0.5, 1)$ and $g_2 = (1, 0.25)$. At the prediction position, $h = (2, -1)$, and P(valve) = 0.25.

The total gradient with respect to the valve row of E is _____. Give a two-component vector.

END OF QUESTION PAPER